

1) Operational modes of a VFD; PU, External, Net and Jog

The operational mode of a variable frequency drive determine the source of the start/stop commands and the frequency selecting the appropriate mode is critical for system integration safety and process automation

i) PU mode:

In this mode, control is localized. The operator uses the physical keypad mounted on VFD to start, stop and manually input the desired frequency. This is primarily used during the commissioning phase localized maintenance tasks where the operator needs direct, immediate control over the motor without relying on external wiring (or) network protocols.

Operator



VFD keypad (PU) ← Controlling here.



Motor

using (or) pressing po/ext button on the VFD keypad

example step

Press Run mode

Set frequency

Press Run

ii) External mode:

The VFD is controlled by external devices

not by the keypad.

ex: Push button, PLC, sensors, etc..

start/stop

VFD

external mode

motor

Start - VFD Terminal

Stop - VFD Terminal

Speed knob - VFD Analogue

ex:

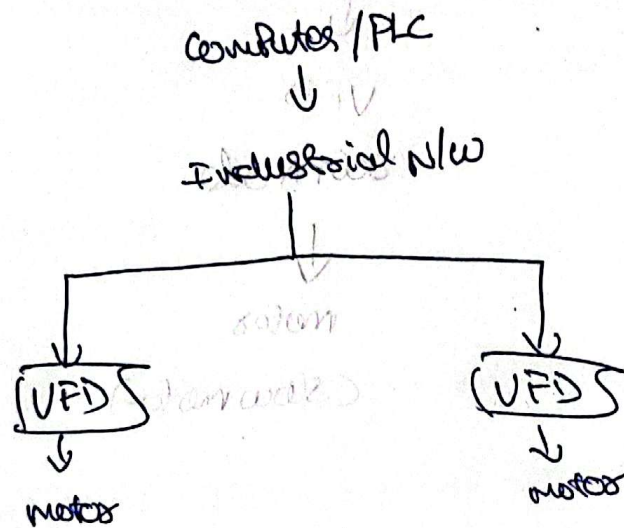
→ Press ext. mode on VFD

→ Control signal from external terminal.

iii) Net mode.

the VFD is controlled by a communication network
ex: mod bus, Ethernet, Industrial network.

Instead of going to each VFD, a centre computer controls them all.



Steps:

Set VFD to NET mode
Connect communication cable
PLC/Computer send command.

Ex:

Run
Stop
Set
Speed.

iv) JCR mode.

means scans the motor for a very short time at low speed. It is used for small speed adjustment.

JCR Button

↓
VFD

JCR mode

↓

motor

(Slow motor)

Steps:

Press JCR Button

motor runs at low speed

Release button → motor stops

Ex:

Normal speed : 50 mg

Low speed : 5 mg

2) Autotuning in VFD : online vs offline processes.

Autotuning is the process by which a VFD identifies

the internal electrical characteristics of the connected motor - such as stator resistance (R_s), rotor resistance (R_r) and leakage inductance (L_s) - to optimize the internal mathematical model used for vector control.

1) offline Autotuning

working:

- * The motor does not rotate during tuning
- * VFD injects a small test signal into the motor windings
- * Electrical Parameters are measured.

Features:

- * Safe for system where motor movement is not allowed
- * Faster tuning process.

* Static Autotuning:

Performed while the motor shaft is stationary. The VFD injects high-frequency signal or DC Pulses to measure resistance and inductance. This is used when the motor is already coupled to a load that cannot be disconnected.

ex:

A conveyor belt system or a packaging plant is connected to a new motor. Before running the system the engineer performs offline auto-tuning to allow VFD to allow the VFD to learn the motor parameters without rotating the conveyor.

ii) online Auto-tuning.

This process occurs continuously while

the motor is running under normal load conditions. As motors heat up, their internal resistance changes.

Online autotuning uses sophisticated observers to monitor the voltage/current vector and adjust the motor parameters in real time to compensate for temperature fluctuation and magnetic flux changes.

3) Braking methods in VFD's : Dynamic vs regenerative.

When a VFD-controlled motor needs to slowdown faster than its natural coast to stop time, when it is being overhauled by a load. the motor acts as a generator. This energy must be managed to prevent an overvoltage fault on the VFD's DC bus.

1) Dynamic braking:

This method dissipates the generated energy as heat. When the DC bus voltage exceeds a set threshold, Braking transistor activates diverting the excess energy into an external Braking resistor.

Working Principle:

It follows the principle of $P = I^2 R$ the kinetic energy of the load is converted into electrical energy and then thermal energy.

working:

- 1) motor adapts during tuning
- * 1) VFD measures dynamic parameters such as inertia and load characteristics.

Features:

- * Provides more accurate tuning
- * used when precise vector control is required

ex:

- * In a CNC machine spindle, precise torque control is required
- * online autotuning is performed while the motor

rotates to obtain accurate dynamic parameters.

ii) Regenerative Braking:

Instead of wasting energy as heat regenerative drives use an Active Front End (AFE) a regenerative unit to convert the DC bus energy back into AC and feed it back into the main power grid.

Working Principle:

It utilizes bi-directional power electronics on the input side of the VFD to synchronize with the line frequency and push energy back to the source.

uses:

* This is used in high-duty cycle application

where significant energy can be recovered. such as

large cranes, elevators, down hill conveyor and

test stands.

4) Classification of Load Patterns and VFD Control modes.

i) Constant torque load:

* The torque required remains constant regardless of the speed. Power increases linearly with speed.

* These loads often require high starting torque to overcome static friction. VFD provides excellent torque regulation at low speed without requiring an encoder, ensuring the motor doesn't stall when starting under full load.

ex: Conveyors, Pumps.

ii) Variable torque loads:

* Torque varies with the square of the speed, $T \propto \omega^2$, and power varies with the cube of the speed.

$$P \propto \omega^3$$

Ex:

Centrifugal fans and Centrifugal Pumps

iii) Constant Power loads

* Torque decreases as speed increases, keeping the product of the two constant
These typically operates in the field weakening region above the motor base speed

=> closed loop vector control

* These application require precise speed and tension control.

* Using an encoder allows the VFD to maintain a very tight and speed loop and manage the motor flux accurately during field weakening to prevent pull out torque issues.